

## 1. Declaration

I, Mandeep Dang, declare that this assignment, titled A1.3: Basic App Dev V2, is my own original work and has not been copied from any other source except where explicitly acknowledged. I have not engaged in plagiarism, collusion, or any other form of academic misconduct in the preparation and submission of this assignment. All sources of information and data used in this assignment have been properly cited and referenced in accordance with the prescribed guidelines. I have not used unauthorized assistance in the preparation of this assignment and have not allowed any other student to copy my work. I am aware that any breach of academic integrity may result in disciplinary action as per the [policies of Monash University](#), which may include failing this assignment or the course, and further academic penalties.

Signature: *Mandeep Dang*

Date: 14/09/2025

## 2. Github Check

Enter your Github details here.

|                                                                                                          |                                                                                                                             |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Github Username<br><i>Enter your username here</i>                                                       | MandeepS2001                                                                                                                |
| Repository Shared?<br><i>Have you started and shared your assignment repository with your tutor yet?</i> | <a href="https://github.com/MandeepS2001/FIT5032-WellManConnect">https://github.com/MandeepS2001/FIT5032-WellManConnect</a> |

## 3. Self-Evaluation

Rate your performance for each criteria. Put a ☒ (tick) in the box where you think your work belongs.

| Criteria                            | Exceeds Expectations                | Meets Expectations                  | Needs Improvement | Fail to meet expectations |
|-------------------------------------|-------------------------------------|-------------------------------------|-------------------|---------------------------|
| BR (C.1): Authentication            | <input checked="" type="checkbox"/> |                                     |                   |                           |
| BR (C.2): Role-based authentication | <input checked="" type="checkbox"/> |                                     |                   |                           |
| BR (C.3): Rating                    | <input checked="" type="checkbox"/> |                                     |                   |                           |
| BR (C.4): Security                  |                                     | <input checked="" type="checkbox"/> |                   |                           |

## 4. Screen Recording of BRs

Create a 3 minute video showing your basic web application in action! Upload this video to your Google Drive and put the link here (ensuring that you have updated the access list so its not private).

<Link to Google Drive Video>

[https://drive.google.com/file/d/1b-10hlCD4eqf68Et4zLN0CV0F2EHK3-i/view?usp=drive\\_link](https://drive.google.com/file/d/1b-10hlCD4eqf68Et4zLN0CV0F2EHK3-i/view?usp=drive_link)

## 5. Reflections: Implementation of C.4 Security

If you have implemented BR C.4, in less than 200 words describe the approach that you have taken to implementing Security in your application. What security flaws were you trying to prevent and what security measures have you implemented to fix those flaws? How do you know that these measures will help prevent those issues from happening? Optionally you can cite external sources to provide evidence for your claim.

In implementing Business Requirement C.4, I focused on addressing common web vulnerabilities such as weak passwords, cross-site scripting (XSS), cross-site request forgery (CSRF), and unauthorized access. For authentication, I enforced strong password requirements and stored them securely using hashing and salting, which prevents plain text exposure even if the database is compromised. To reduce the risk of XSS, I validated and sanitized user input before displaying it, ensuring that malicious scripts could not be executed in the browser. I also added token-based CSRF protection so that only valid requests made within a user's session are accepted. In addition, I implemented role-based access control so that only administrators can access sensitive areas like the admin panel.

I believe these measures are effective because they align with best practices such as the OWASP guidelines, which are designed to defend against the most common and damaging attacks. By following these standards, I feel confident that the application is better protected, and I have gained a stronger understanding of how layered security creates resilience against multiple types of threats.

## 6. Reflections: Challenges

What has been the most challenging part of this assignment for you? How has this stretched you as a programmer?

The most challenging part of this assignment was implementing security features in a way that went beyond simply making the application "work." I realized that it was not enough to just build functionality — I had to think like an attacker, anticipate vulnerabilities such as XSS or CSRF, and then design effective protections. This required me to step outside my comfort zone and learn about input sanitization, token-based validation, and secure password hashing. It also stretched me as a programmer because I had to balance technical implementation with usability; for example, enforcing strict password rules improved security but risked frustrating users. Another challenge was managing my time while debugging errors that came from small mistakes in validation logic. These moments

tested my patience but also strengthened my problem-solving skills. Overall, the assignment pushed me to think more critically about secure design, gave me practical experience with industry best practices, and helped me grow confidence in building applications that are both user-friendly and resilient against real-world threats.

## 7. Declaration: Additional Help

Any tools that you used (including Gen AI or existing code reuse) must be declared here.

**Note:** GenAI is not allowed for coding purposes in any assignment,

However, you may use GenAI for brainstorming and problem solving. You need to declare all such uses here. One row per help used.

| Name           | Description                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>ChatGPT</i> | <i>I used ChatGPT to brainstorm how to approach different parts of the assignment because I was feeling stuck on how to organize my ideas and present them clearly</i> |
|                |                                                                                                                                                                        |